International Finance 4832
Lecture 7: Open Economy Macroeconomic Policy in the Short Run

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Currency market

IS-LM - FX

goods market, money market

Analysis similar to before, but Y is no longer constant instead it is affected by ER, TB

Outline

Before:

1. (Accounting) Measure of International Transactions (BOP) and wealth (NFA)
2. CA $\neq 0 \Rightarrow$ borrowing/lending with ROW
3. The Long Run Budget Constraint (LRBC): PV of CA flows = - PV of wealth
4. Gains from financial globalization (international lending/borrowing)

   Consumption smoothing, efficient investment, risk diversification

Now: (Chapter 18) Policy Framework for analyzing the Open Economy

Open Economy IS-LM (or IS-LM-FX model)

The model links: Money market, FX market and Goods market equilibria

Next: Part 3, Economic Policy Applications

Where we left

Now we use what we studied on the FX market and the Exchange Rate (in short run)

Foreign Exchange market $\longrightarrow$ modeled via UIP → Domestic returns ≈ Foreign returns

Money market $\longrightarrow$ modeled with the quantity theory of money

Before: we assumed output was constant (Y) (e.g., in $M/P = L(i)Y$)

Now we include the **Goods Market** into the model ... allowing Y to change

We consider a short run model $\longrightarrow$ Prices are fixed

IS-LM model setup extended to work for Open Economies → adds TB & Exch. Rate to the model (E_{hlf}, TB)

Focus: home country vs. ROW

The model considers each element of Y: $Y = C + I + G + TB$

Simple long run budget constraint

Some simplifying assumptions

1. Short run: prices are fixed, inflation is zero

   (thus we can speak about real and nominal quantities as if they were the same)

2. Government policy is constant ($\bar{G},\ \bar{T}$)

   later we allow changes in Fiscal Policy

3. Foreign output and interest rate are both constant and exogenous: $\bar{Y}^*,\ \bar{i}^*$

4. $NFIA = 0,\ NUT = 0$, then: $CA = TB$

Now, let's look at each of the GDP's components ...

Consumption

"C" in $Y = C + I + G + TB$

Model consumption as a function of (disposable) income:

$$C = C(Y - \bar{T})$$

Notice the parenthesis here denotes $C()$ is a function, and not C times $Y - T$

This function $C()$ increases in disposable income $Y - \bar{T}$

Slope of consumption: marginal propensity to consumption (MPC)

MPC + MPS = 1

$0 < MPC < 1$

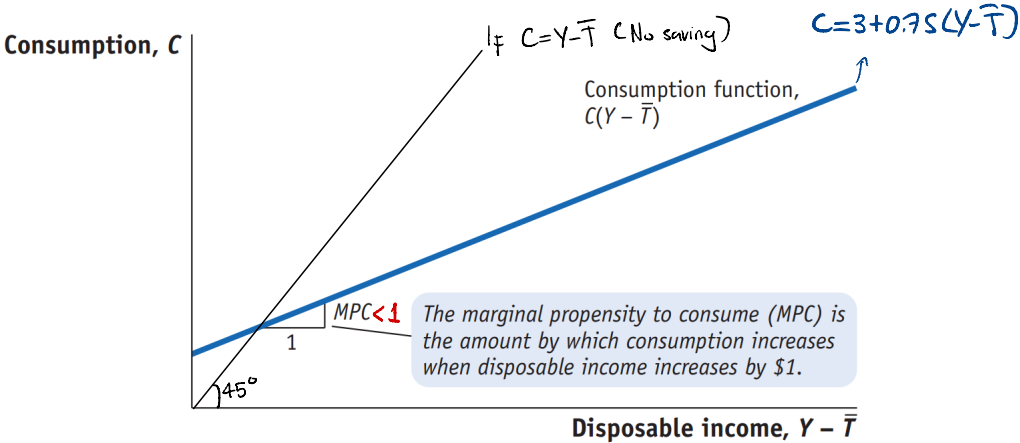
Example:

$$C = 3 + \underbrace{0.75}_{MPC}(Y - \bar{T})$$

Here: $MPC = 0.75$, i.e., for any extra dollar of disposable income the households spend 75 cents

Consumption function

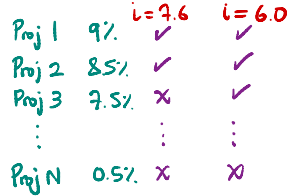
Figure: Consumption



Investment

"I" in $Y = C(Y - T) + I + G + TB$

Many possible investment projects, each paying a different real return
(e.g., Google, open a food truck, buy a farm, etc)



Projects whose rate of return is greater than expected real interest will be undertaken

$$r_{\text{project}} > r$$
$$r_{\text{project}} > i - \pi^e$$

If $\downarrow i \Rightarrow$ Invest in more projects

$\Rightarrow \uparrow$ Aggregate Investment

In the short run $\pi^e = 0$ (sticky prices), then:

$$r_{\text{project}} > i$$

i.e., we just need the real return of projects to be equal or greater than nominal interest rates

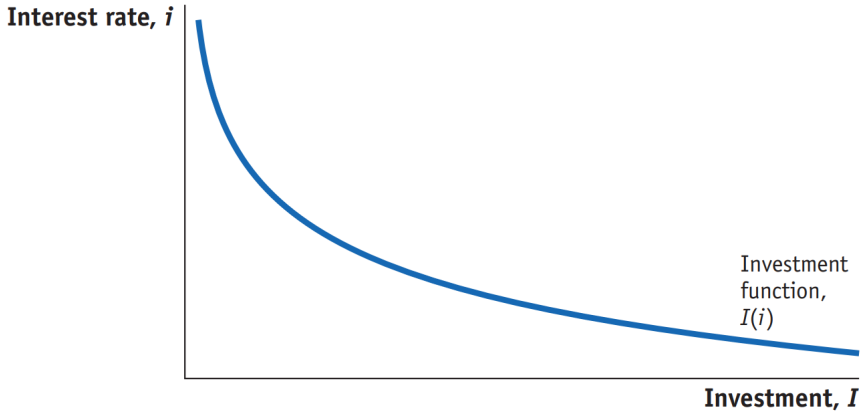
Negative relationship between the aggregate investment and the interest rate:
The lower is i the more projects will be undertaken $\longrightarrow$ the higher the total investment (I)

Total investment: I $\longrightarrow$ decreasing function of i

$$I = I(i)$$

Investment function

Figure: Investment



Government

"T" and "G" in $Y = C(Y - T) + I + G + TB$

Revenue for Gov.

Spending of Gov.

Revenue: Collects taxes T

Expenditure: Purchases of public consumption goods G

T, G are assumed exogenous and fixed (do not depend systematically on other factors in the model)

To denote this explicitly we use bars: $\bar{T}$, $\bar{G}$

Link with the households: for households the disposable income will be the remainder after paying taxes: $Y - \bar{T}$

We later allow for **Fiscal Policy**: that is, changes in G, T

These quantities will change at the discretion of the government and policymakers ...

...and not as a response to other variables (⇒ still assume these variables don't depend on model's features)

Trade Balance

"TB" in $Y = C(Y - T) + I + G + TB$

↳ or "NX" ($NX = EX - IM$)

New in the model relative to previous courses

Drivers of Trade Balance:

1. Prices: Real exchange rate (EP^*/P)
2. Incomes: home and foreign (the higher, the more demand of each location for imports) (Y, Y^*)

That is:

$$TB = TB(EP^*/P, Y - \bar{T}, Y^* - \bar{T}^*)$$

(+) (−) (+)

TB *increases* in EP^*/P

TB *decreases* in $Y - \bar{T}$

TB *increases* in $Y^* - \bar{T}^*$

Furthermore, with fixed prices we can see that TB increases in E too (i.e., in nominal ER too)

Let's look at each of these components driving the TB ...

Trade Balance and the Real Exchange Rate

Real Exchange Rate:

$$q_h = \frac{E_{h/f} P_f}{P_h}$$

In the long-run, by PPP if follows that $q_h = 1$

In the short-run, prices are fixed and changes in $E_{h/f}$ lead to proportional changes in q_h

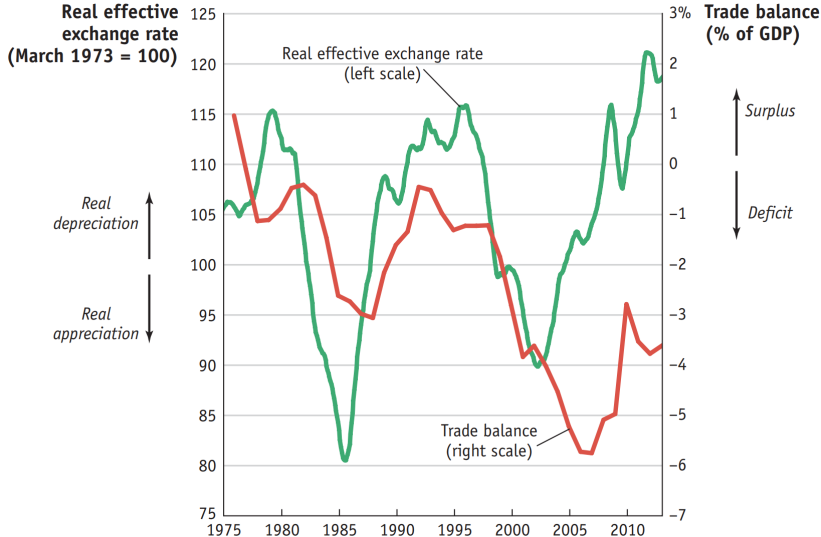
(we can think of the Real and Nominal ER as analogous variables only due to sticky prices)

If $E_{h/f}$ **falls** $\longrightarrow$ foreign goods become cheaper relative to home goods $\longrightarrow$ buy more foreign, sell less home goods (import more, export less) $\longrightarrow$ **$\downarrow$ TB**

If $E_{h/f}$ **rises** $\longrightarrow$ foreign goods become expensive relative home goods $\longrightarrow$ buy less foreign, sell more home goods (import less, export more) $\longrightarrow$ **$\uparrow$ TB**

Therefore: TB is increasing in the Exchange Rate (real or nominal)

Trade Balance and Real Exchange Rate in the US



Source: FRED

Trade Balance and Income levels

An increase in **disposable income** leads to higher consumption

This is true in any location: home or foreign …

…and for any kind of goods consumed, including *imports*

Then:

- An increase in home disposable income $(Y - \bar{T})$ leads to an increase in *home imports* $\rightarrow \downarrow$ TB
- A higher foreign disposable income $(Y^* - \bar{T}^*)$ leads to an increase in *home exports* $\rightarrow \uparrow$ TB



What about the taxes? $\longrightarrow$ link between fiscal policy and trade balance
(We'll see the overall effect later)

Shocks to C, I and TB

Factors that change C, I, G, TB will shift the Aggregate Demand

Here are some examples of such Shocks:

Shocks to C: Households consume more (or less) for any level of income

Example: ↑ in wealth from higher home prices or increase in savings due to a possible natural disaster

Shocks to I: Firms invest more (or less) for any given interest rate

Example: Technological discovery such that new and more profitable projects become available

Shocks to TB: Households consume more (or less) foreign goods relative to home goods

Example: Change in tastes towards foreign goods

As usual, any sudden increase (or decrease) in C, I, G, TB for the same level of output generates the same change in the Aggregate Demand

(and will imply shifts in its curve once represented in a plot)

Fiscal policy: when we allow changes in G and T these also represent a type of shock (that shift demand) that are induced by the government at their discretion

The Equilibrium in all markets: IS, LM, FX

Goods market equilibrium

We have functions for C, I, TB (and we know these are the possible types of goods we consider)

T, and G: exogenous

Fixed for now and change only at the discretion of the government and not by reacting to other changes in the model like the other variables (e.g., C, TB)

$CA = TB$ (given assumptions on $NFIA$, NUT)

Then:

$$\text{Supply of goods} = Y,$$
$$\text{Demand of goods} = C + I + G + TB,$$

New

In equilibrium:

$$\text{Demand} = \text{Supply}.$$

That is:

$$Y = C + I + G + TB.$$

In terms of the functions we defined:

$$Y = C(Y - \bar{T}) + I(i) + \bar{G} + TB(E\bar{P}^*/\bar{P}, Y - \bar{T}, Y^* - \bar{Y}^*).$$

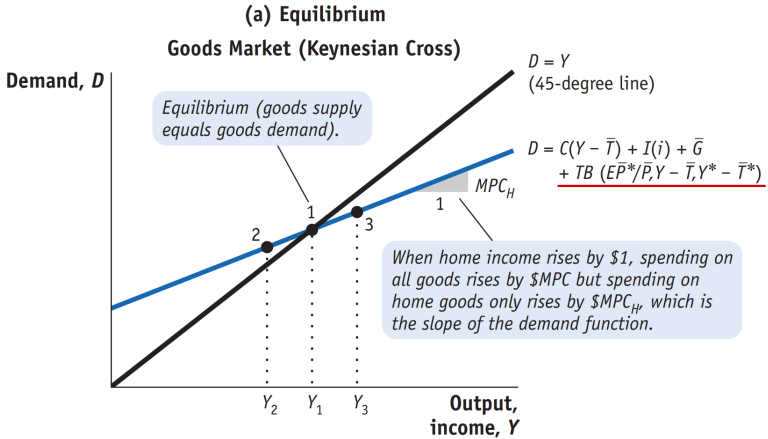
link w FX markets (WIP)

Keynesian Cross

Typical plot representation of this equilibrium from previous macro courses: The Keynesian cross

Keynesian Cross: Demand vs. Supply with equilibrium at intersection where Demand $=$ Supply

Figure: Goods Market (Equilibrium at intersection)



Slope of Demand

Positive Slope: Demand Slopes up $\longrightarrow$ see how each component grows (or not) with income:

Income shows up in C and TB (i.e., $Y - \bar{T}$ in each function)

$\uparrow$ C by $MPC > 0$

$\downarrow$ TB though: Imports increase by $MPC_f \Rightarrow$ Demand decreases by -MPC_f

Total effect on Demand: $MPC - MPC_f = MPC_H > 0$

If more of the extra income goes to domestic goods than foreign goods: Demand increases

(this is usually the case ... $MPC - MPC_f = MPC_h > 0$)

Second: Slope is lower than 1

Each \$1 will increase demand **for home goods** less \$1 and than MPC ...some goes to imports

So actual increase is $MPC_h = MPC - MPC_f \rightarrow$ lower than $MPC < 1$ (with savings > 0)

(note: Slope < 1, i.e., of MPC_h implies a single intersection –or equilibrium– between supply and demand in the cross)

Equilibrium in Goods Market

The equilibrium output is described by $Y = C + I + G + TB$:

If actual output is *lower* than the equilibrium one $\longrightarrow$ Firms produce *more*

Shortage of production

If actual output is *higher* than the equilibrium one $\longrightarrow$ Firms produce *less*

Surplus of production

Eventually, the equilibrium and actual output coincide:

$Y = D$, and $D = C + I + G + TB$ (the curves intercept)

Demand driven equilibrium: output adjusts to meet demand in the short-run (with sticky prices)

In the long-run prices are flexible and adjust to clear the market

In a nutshell:

SR: Adjustment via quantities LR: Adjustment via prices

Shifting Demand

Private consumption increases after:

1. A decrease in taxes ↓T
2. Other factors affecting consumption but not income (e.g., tastes)

That is, when we have a higher C for any level of Y (it shifts up)

Investment increases after:

1. A decrease in i ↓i
2. Other factors driving up investment but not income (e.g., technological discovery)

That is, when we have a higher I for any level of Y (it shifts up)

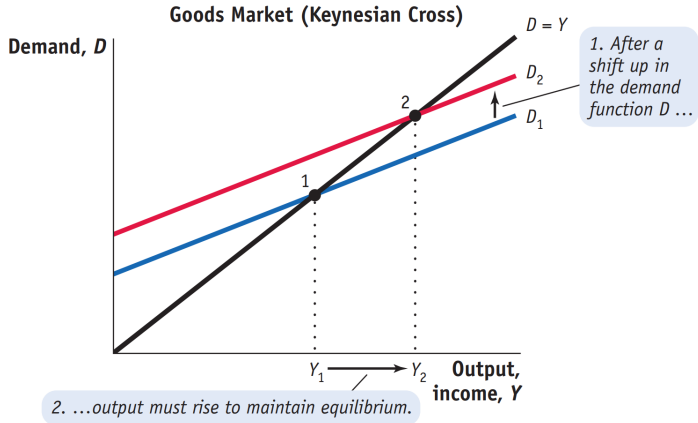
Trade balance increases after:

1. An increase in Real ER: ↑ E (depreciation) (with sticky prices ↑ E is the same as ↑ RER)
2. An increase in P^* or a decrease in P
3. An increase in T or a decrease in T^*
4. Other factors affecting the TB but not the income

That is, when we have a higher TB for any level of Y (it shifts up)

Goods market equilibrium: Increase in demand

Figure: Goods Market (Equilibrium at intersection)



Shift up (down): higher (lower) demand for any level of income

General Equilibrium



Three markets: Goods, FX (forex), Money

All of them should clear simultaneously (be in equilibrium)

We connect the markets with 2 extra plots:

- IS curve $\longrightarrow$ links goods and FX markets
- LM curve $\longrightarrow$ links goods and money markets

Let's derive each now, then we can put everything together later

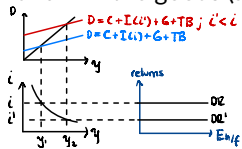
IS curve

Relates the goods market to the Forex market $\longrightarrow$ Link: interest rate i

Recap: In the goods market shows up via I, in the FX market shows up in UIP (DR and FR curves)

IS: represents all the $(i,\ Y)$ pairs that are consistent with equilibrium in the goods (and FX market)

Derived: by changing i and tracing out the corresponding Y



New result: (a decrease in rates) $\downarrow i$ stimulates economic activity **more** in Open Economies

- Closed economy effect on Investment works the same way
- But it also causes a depreciation and increases the trade balance (TB)

IS curve slopes down:

$\downarrow i$ then $\uparrow I, \uparrow E_{h/f}$ (depreciation of ER) and $\uparrow TB$ ...from both effects: $\uparrow Y$

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↳ New: TB reacts to change in i because of $E_{h/f}$

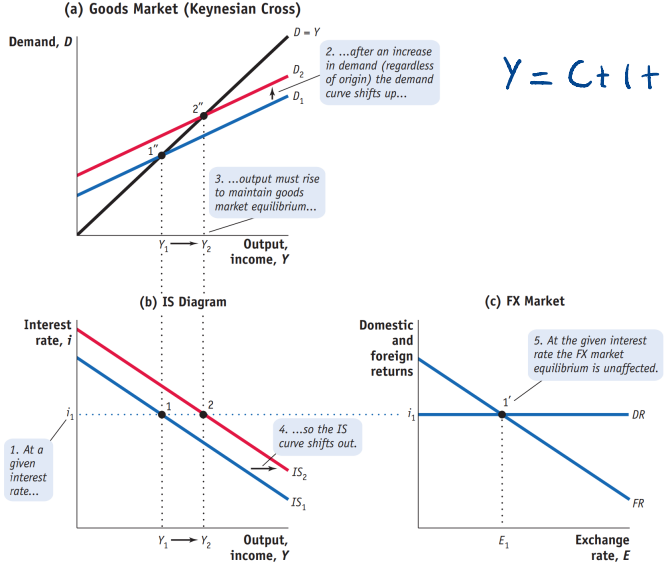
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$\downarrow i$ then $\uparrow I, \uparrow E_{h/f}$ (depreciation of ER) and $\uparrow TB$ …from both effects: $\uparrow Y$

($\downarrow i \xrightarrow{UIP} \uparrow E_{h/f} \rightarrow TB\uparrow$; $Y\uparrow$)

Shifting the IS curve

Figure: Shift in IS curve after an Increase in G



$$Y = C + I + G + TB$$

LM curve

Represents all the $(i,\ Y)$ pairs such that the money market is in equilibrium

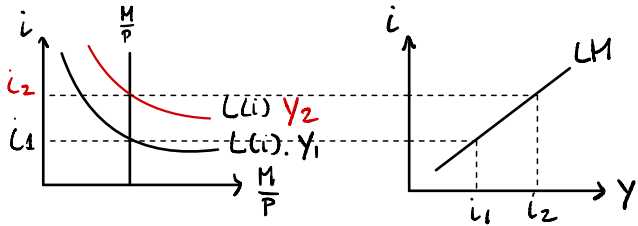
Derived it by changing Y and tracing out corresponding i

(Note: Same setup as in a closed economy)

(Unlike the IS) the LM curve setup is the same as in a closed economy

The **LM** curve as an **upward sloping curve**:

If real output rises people demand more money, with money supply fixed interest rate rises: $\uparrow Y \longrightarrow \uparrow i$



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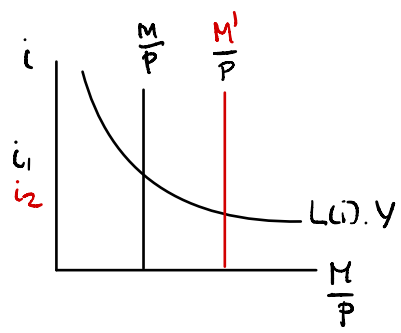
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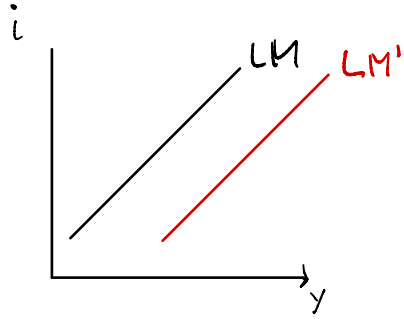
Shifting the LM curve

Increase in (nominal) money supply

Things that shift money demand down that are not Y

Example: increase in $M \longrightarrow \uparrow \frac{M}{P}$





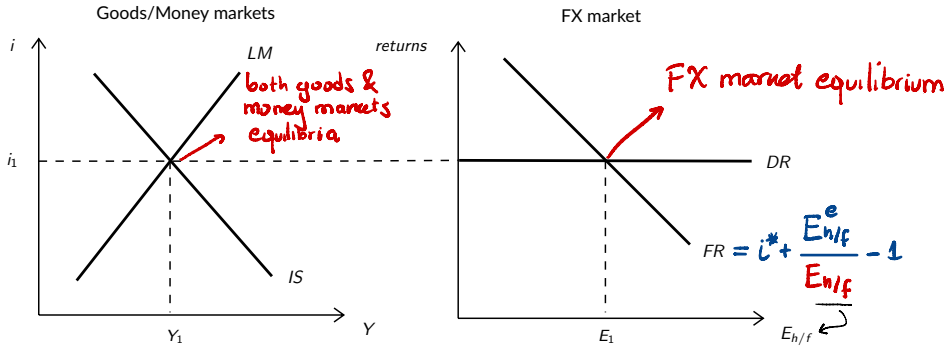
IS-LM-FX

Points along the IS curve: combinations of $(i,\ Y)$ where goods and FX market are in equilibrium

Points along the LM curve: combinations of $(i,\ Y)$ where money market is in equilibrium

Point shared by IS and LM (**intersections**) $\longrightarrow$ all 3 markets are in equilibrium

Determining $i,\ Y,\ E$ equilibrium values



Summary

New in open economies

IS: Combinations of $(i,\ Y)$ such that goods and FX markets are in equilibrium

- $\downarrow T, \downarrow P$, or $\uparrow G, \uparrow i^*, \uparrow E^e, \uparrow P^*$ shift demand (up) and IS (right)
- Any change in $C(\cdot)$, $I(\cdot)$, or $TB(\cdot)$ shifts demand, shifts IS

  i.e., important factors that are not Y

LM Combinations of $(i,\ Y)$ such that money market is in equilibrium

- $\uparrow M$ shifts LM down (or to the right)
- Any change in money demand function that shifts money demand down $\longrightarrow$ shifts LM down

  (i.e., changes in $L(i)$ but not in Y)

→ Use the UIP to determine $E_{h/f}$

The Short Run IS-LM-FX model of an Open Economy

↳ Similar to a closed economy

But w/ a TB that depends on $E_{h/f}$

Outline

Before:

1. IS-LM-FX ties together equilibrium of Forex, money, and goods markets
2. IS-LM-FX will be our workshorse open economy macro model (for most topics ahead)

Now: IS-LM-FX and policy

1. Fiscal policy in an open economy
2. Monetary policy in an open economy
3. Effects of policy under different (ER) exchange rate regimes



IS-LM-FX recap

IS-LM intersection:

- Goods and FX market are in equilibrium (IS curve)
- Money market is in equilibrium (LM curve)

IS curve shifts:

- $\downarrow T, \downarrow P$, or $\uparrow G, \uparrow i^*, \uparrow E^e, \uparrow P^*$ shift demand (up) and IS (right)
- Any change in $C(\cdot)$, $I(\cdot)$, or $TB(\cdot)$ shifts demand, shifts IS

  i.e., important factors that are not Y

Related to "FR" in the UIP

LM curve shifts:

- $\uparrow M$ shifts LM down (or to the right)
- Any change in money demand function that shifts money demand down $\longrightarrow$ shifts LM down

  (i.e., changes in $L(i)$ but not in Y)

Policy in an Open Economy

Monetary and Fiscal policy

Assumptions

1. Policy changes are temporary: Do not change expectations (e.g. E^e, P^e)
2. Short-run analysis: Prices are sticky
3. Free movement of capital: UIP holds
4. Variables in foreign country are given (and fixed)

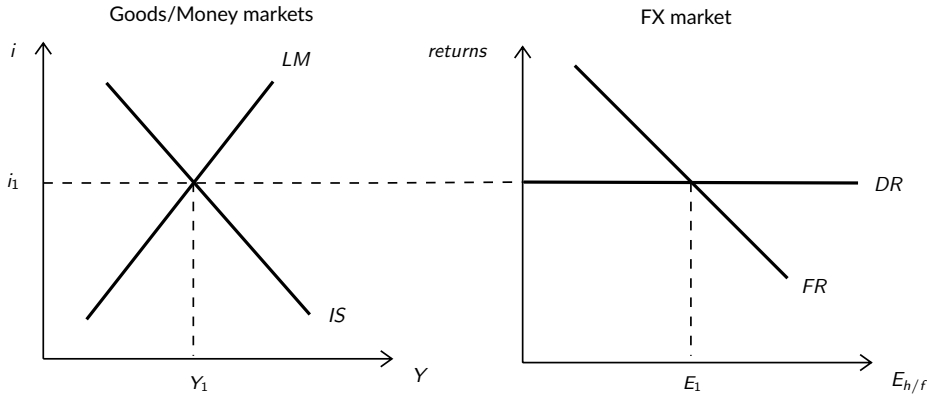
Temporary policies are a more interesting case than permanent ones

Resembling of policymakers responding to cyclical conditions and shocks (more realistic)

Start with flexible exchange rates (→ Floating ER) then look at the Fixed ER regime case (→ ER peg)

We will see how the regime choice (constrains) changes the effects of the policies

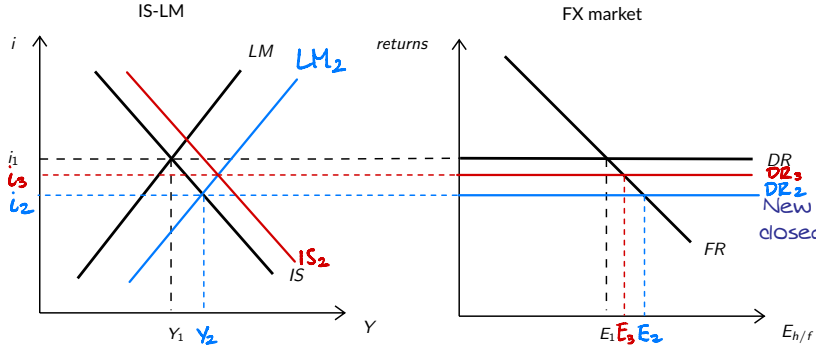
Initial Equilibrium in IS-LM-FX



Monetary policy: flexible exchange rate

$E_{h/f}$ can change

Central bank (temporarily) expands money supply (↑ M) … Does IS or LM shift? Where?



New with respect to closed economy IS-LM

Drop in i shifts down DR curve (in FX market via UIP) … depreciating the ER (↑ E)

Takeaway: In closed economy ↑ Y due to ↑ I; Open economies: Too but rises further due to ↑ TB

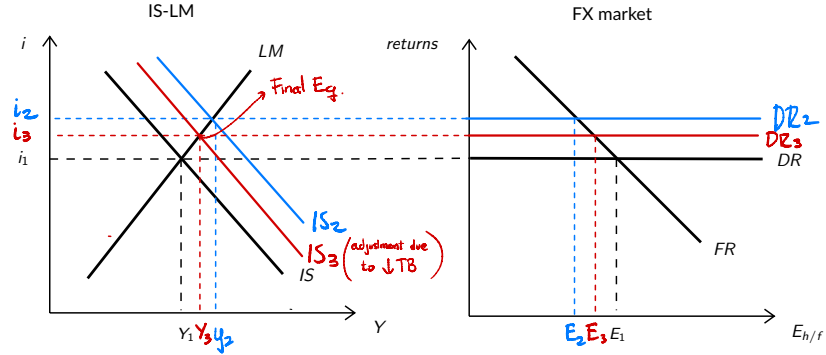
Food for thought: ↑ M increases Y as we saw … this also increases *imports* which would lower TB

⇒ TB rises due to ↑ E and lowers due to ↑ imports … Assumption: overall effect on TB is positive
Known as the Marshall Lerner effect (familiar?) (overall effect of depreciation on TB is positive, TB rises)

Fiscal Policy: flexible exchange rate

Government temporarily lowers taxes ($\downarrow T$) ... Does IS or LM shift? Where?

Final Eq: i increases
Y increases
E appreciates



IS shifts right due to $\uparrow C(Y - T) \rightarrow \uparrow$ interest rate and decreases I $\Rightarrow$ Overall effect: Y increases

New with respect to closed economy: $\uparrow i$ appreciates E lowering the trade balance $\longrightarrow \downarrow TB$
(i.e., the TB is partly crowded out)

These extra effects (on I, TB) partly explain why with higher G, Y rises but less than proportionally (fiscal multiplier)

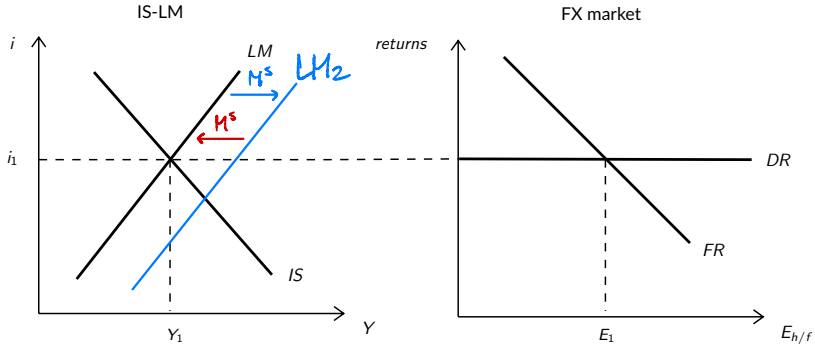
Monetary policy: fixed exchange rate

$E_{h/f}$ cannot change

Not possible to conduct monetary policy $\longrightarrow$ trilemma in action

Fixed exchange rate & free capital mobility $\Longrightarrow$ no monetary policy

Even if done: Monetary policy changes must be undone to keep i constant and the E fixed



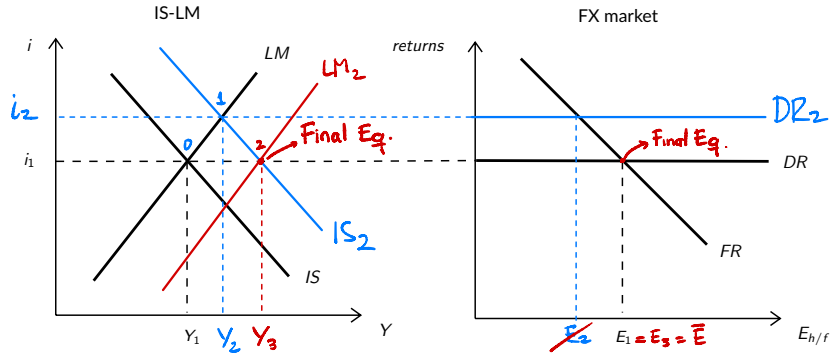
Monetary expansion would depreciate E which is not consistent with the peg regime

no change in equilibrium $\longrightarrow$ $\uparrow M^s$ not possible here (in absence of other shocks)

Fiscal Policy: fixed exchange rate → $E_{h/f}$ cannot change

Decrease in taxes, how to IS and LM shift? Does LM have to shift?

↓T
(↑C)



Notice: Fiscal policy becomes super effective in boosting Y (Y ↑ by more than without a fixed ER)

Why? ⟶ crowding out effects (on I, TB due to changes in i are removed)

Central bank policies: reduced to undoing effects of Fiscal Policy on i (to defend the peg)

Policy implementation: Effectiveness and predictability

These policies have practical effects, many in the direction we have discussed … but sometimes the outcomes may differ from what was planned

In real life some factors difficult the planning:

Policy constraints $\longrightarrow$ difficulty to raise funds for fiscal policy

Inside lag: time lag between shocks an policy actions

Outside lag: time lag between policy actions and effects

Incomplete Information: governments may not have full picture of state of the economy in time

Long horizon plans: agents may act based on long investment/consumption plans and base decisions on expectations and not on temporary ER, i changes

Incomplete RER to NER passthrough

Intermediate ER regimes (e.g. managed floats)

Weak link between ER and TB: Trade costs in trade (that were assumed 0) and J curve effects (coupled with trade costs may even lead to contrary response of TB to ER movements)

Summary

Responses to Policy Shocks in the IS-LM-FX Model

Exchange Rate Regime	Policy	Impact on: i	E	I	TB	Y
Floating	Monetary expansion	↓	↑	↑	↑?	↑
	Fiscal expansion	↑	↓	↓	↓	↑
Fixed	Monetary expansion	0	0	0	0	0
	Fiscal expansion	0	0	0	↓	↑

Crowding out

No MP if E fixed

Super effective Fiscal Policy

Effects in Open Economies:

Additional to usual (closed economy IS-LM) effects: We must account by movements in ER & TB

Changes in i lead to changes in the $E_{h/f}$ and TB that add to the overall effect on Output (Y)

Policy effects depend on the FX regime:

Counterbalancing effect on expansionary policies **if they imply a higher interest rate**:

They prompt a crowding out of Investment and a ER appreciation a with lower TB

In Fixed ER regime: Crowding out effects are removed $\longrightarrow$ policy makes sure i does not change

$\Rightarrow$ Expansionary fiscal policies: more effective ... At what cost? $\longrightarrow$ lose monetary policy usage

Stabilization Policy

Outline

Before: IS-LM-FX model

1. Fiscal and Monetary policy with Floating ER
2. Fiscal ~~and Monetary~~ Policy with Fixed ER

   Central Bank only accommodates Fiscal Policy to ensure ER stays fixed in this case

Now: Stabilization policy

Application of IS-LM-FX model

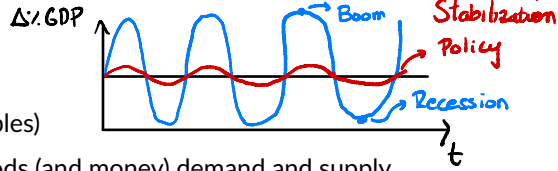
Next: Part 3, Economic Policy Applications

1. Fixed vs. Floating ER regimes in more detail
2. Historical context: Gold Standard, Bretton Woods, ERM
3. Exchange rate crises and models

Stabilization Policy

Economies are constantly affected by "shocks" $\longrightarrow$ shocks: exogenous changes

- Changes in technology, regulation
- Market bubbles bursting
- Natural disasters, etc (many more examples)



These shocks change the components of goods (and money) demand and supply

- Shifting IS/LM/FR $\longrightarrow$ leading to a new equilibrium
- By changing equilibrium $\longrightarrow$ output, interest rate and ER can change too

These shocks create recessions and expansions $\longrightarrow$ "Business Cycles"

Stabilization policy: effort to smooth out these shocks

Objective: keep economy at or near full-employment level of output

Facilitating a stable environment for long-term decision making (e.g., of investment)

With smooth income $\longrightarrow$ easier to achieve smooth consumption

Stabilization Policy

Target level of output: "potential" GDP $\longrightarrow$ output associated to "Full employment"

Full employment: highest level of employment that can be sustained without $\uparrow$ inflation

Policymakers want to **minimize deviations** from the target

e.g., by cooling down (expanding) economy if GDP is too high (low)

We saw how shocks are reflected in the IS-LM-FX framework (e.g., higher foreign rate)

Stabilization policy tries to offset shocks that move the GDP from target

Done by implementing a policy change with the opposite effects

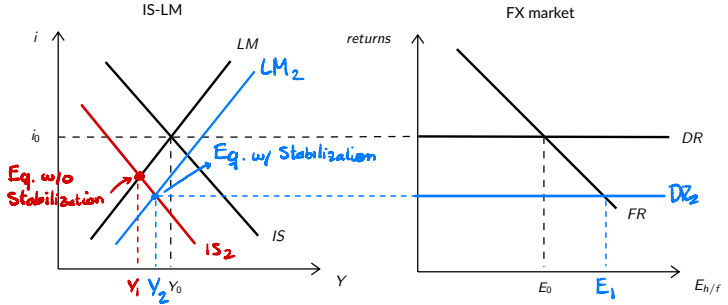
What policies can be used to do this? Fiscal (T,G), Monetary (M, i)?

- Monetary: more flexible and timely $\longrightarrow$ easier to use (and more common)
- Fiscal: used less frequently but available $\longrightarrow$ normally set with longer term goals in mind
- In fixed ER regimes: monetary policy is not an option (must use fiscal)

Stabilization policy in the IS-LM-FX model

1. Country is at target GDP level (Y_0, or $Y^{potential}$)
2. Shock occurs, disrupts IS-LM initial equilibrium and brings output off-target
3. Monetary and Fiscal policy responses
   - Push economy closer to target
   - Response depends on ER regime

Example: negative shock to C, I (e.g., lower consumer and investment confidence)



Additional Implementation Problems for Open Economies

Same problems with implementation as mentioned before: Policy constraints, policy lags, etc.

Additional: In model: ER changes $\longrightarrow$ Real ER changes $\longrightarrow$ Trade Balance changes $\longrightarrow$ IS shifts

In reality any of the connections made above may weaken, even break

Nominal to Real ER link: ($\Delta E \Rightarrow \Delta q \Rightarrow \Delta TB$) P assumed completely fixed $\Rightarrow$ moving $E \approx$ changing q

This is not always true: several factors may weaken link:

- Extent of response of P to E (ER passthrhough)
- Dollarization of trade
- Monopolistic power and pricing to market $\longrightarrow$ firms setting different prices (and markups) in different locations

Real ER to TB link: In reality TB may not always react to ΔE (or Δq) due to additional factors:

- Transaction costs
- Lagged response of imports and exports (e.g., if sales subject to prenegotiated contracts, etc)

Intermediate ER regimes: Many countries work in regimes resembling a mix of a peg and floating.

Stabilization in Reality

Monetary policy:

In model: Shock $\rightarrow$ Policy response: Exact, on time and intended effect

Reality: Shock + (incomplete info + lags + limited efficacy) $\rightarrow$ Policy response: Slow, partially intended effects, unintended effects, hard to assess

Policy rates in the US over time: https://fred.stlouisfed.org/org/graph/?g=m2nm

Fiscal Policy:

- Longer lags + policy efficacy constraints $\rightarrow$ used infrequently
- A lot of "red tape" involved + can be tied up for political reasons
- Difficult to raise and spend money quickly (adding to lags, etc)

Overall:

Monetary policy is more flexible and thus more capable of responding to short-run shocks

Fiscal policy is designed with medium and long-run goals in mind instead

Limits of Monetary Policy: Liquidity Traps

Key to monetary policy: The Interest rate

- $\downarrow M \rightarrow \uparrow i$
- Acts by changing money supply but equilibrium effects happen through interest rate change
- Change M (supply) $\rightarrow$ new interest rate $\rightarrow$ LM curve shifts (same interest rate for any Y level)

Constraint: Nominal interest rate cannot go below zero $\longrightarrow$ "Zero Lower Bound"

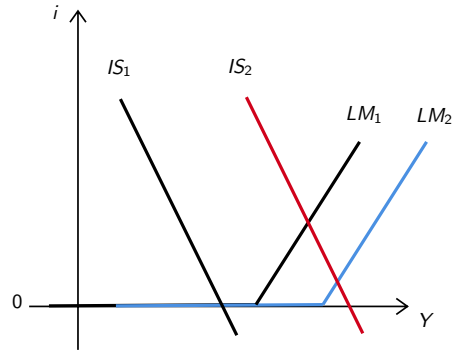
When $i = 0$ monetary policy no longer works $\longrightarrow$ Liquidity trap

Liquidity trap: Injecting more liquidity ($\uparrow M$) in the economy is unable to affect rates
(in IS-LM: looks like a horizontal region in LM curve $\sim$ many levels of output for same i)

A liquidity trap creates an important role for fiscal policy

Expansionary Fiscal Policy becomes the way to push the economy out of the liquidity trap

The liquidity trap in the IS-LM-FX model



Fiscal Policy is Key here:

- Is either super effective as there is no crowding out of I or TB
- Or can be helpful to take the economy out of the liquidity trap

The liquidity trap (cont.)

Two prominent examples:

1. Japan in early 2000s
2. United States (and other countries/regions) in 2008-2021, begginging of 22.

Real estate bubble collapse: decreased wealth

Overleveraged households and business cut back on: C, I

Prompting IS shifts to the left: Interest rates fell to zero

US: https://fred.stlouisfed.org/org/graph/?g=m2nm

Japan: https://fred.stlouisfed.org/series/IR3TCD01JPM156N

The U.S. policy response

During Global Financial Crisis of 2008: American Recovery & Reinvestment Act (ARRA)

- Initially to be $1.4 tril in extra government spending and tax cuts (was $ 787 bil)
- Lags: recession starts in 2007Q4. ARRA signed (law): 2/2009. Bulk of implementation: 2010

ARRA not thought to be very effective: G didn't change by much (or enough)

States cut back on spending about as much as federal government raised it

During Covid-19 lockdown: Coronavirus Aid, Relief & Economic Security Act (CARES)

- $ 2.2 trillion stimulus
- Not very effective (C didn't go up) $\longrightarrow$ people saved the subsidies rather than spending them
- Then further stimulus was extended
- IS shifted although we are seeing the inflationary effects in 2022